Digital Scholarship in the Humanities, 2025, **40**, 295–307
https://doi.org/10.1093/llc/fqae090
Advance access publication 2 January 2025
Full Paper

Analysis of modern strategies for using artificial intelligence technologies in the creation of fantasy content

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Abstract

This study explores the integration of artificial intelligence (AI) in the creation of fantasy narratives, examining the new possibilities AI offers for authors and readers and its impact on the genre's evolution. The study used literature review, textual analysis, comparative analysis, data analysis, experimental methods, ethical and legal examination, and data processing methods using Natural Language Processing software. Through analysis of films, video games, and qualitative text review, the research found that modern technologies—particularly AI and computer graphics—have greatly enhanced the visualization of fantasy worlds, contributing to the genre's growing popularity. Interactive AI-driven platforms enable customized experiences, increasing engagement and satisfaction among audiences. A significant finding is that contemporary fantasy works, while retaining traditional elements like the struggle between good and evil, magic, and adventure, are now incorporating modern socio-cultural and political themes. This blend preserves the genre's appeal to diverse audiences while keeping it relevant in today's context. Additionally, AI opens new avenues for authors by aiding in overcoming creative blocks, generating fresh ideas, and constructing interactive worlds where readers can shape the plot or create their own characters. This not only accelerates the creative process but also enriches storytelling with more varied narratives. The study concludes that fantasy remains a powerful tool for cultural and social expression and that AI holds great potential for the genre's future growth, promising more innovative and inclusive experiences for audiences.

Keywords: mythology; visualization; computer games; socio-cultural changes; interactive platforms; literature.

1. Introduction

The relevance of the study is conditioned by the rapid development of artificial intelligence (AI) technologies and their impact on various fields of creativity, including literature. The introduction of AI into the literary process opens up new horizons for authors, allowing them to speed up the creation of texts, improve their quality, and offer new forms of interaction with content. This research is important because it helps to understand how AI can change and enrich the fantasy genre, making it more accessible and diverse. The main problem of the study is related to a number of challenges and limitations that arise when integrating AI into literary creativity. Ethical and legal aspects include copyright, intellectual property, and liability issues when using AI to create literary works. The quality and creativity of texts raise concerns that AI can completely replace human creativity, and problems related to ensuring high quality and creativity of generated texts. Technical limitations lie in the fact that modern AI algorithms have not yet reached the level that allows creating completely autonomous and creative works without human intervention. The research aims to overcome these challenges and find ways to effectively integrate AI into the process of creating fantasy literature, which can lead to significant innovations in this area (Freeman and Smith 2023a).

Machine translation systems as artificial intelligence tools are still at the stage of improvement, despite their powerful base of lexicographic sources of various types (reference books, dictionaries), which still require greater involvement of human resources to achieve adequate text translation (Etemi and Uzunboylu 2020). This issue was investigated by Kapranov and Glushchenko (2023), who highlighted the limitations of current machine translation systems and the necessity of human intervention for accurate and contextually appropriate translations. However, they did not consider the potential impact of neural network technologies on improving the quality of translation, which could offer more nuanced and context-aware translations by learning from vast amounts of data.

In the field of artificial intelligence image classification in science fiction, Osawa *et al.* (2022) note that this helps to communicate more clearly by identifying elements with which their work may or may not be similar. Their study emphasized the role of AI in enhancing visual storytelling and aiding creators in maintaining consistency and clarity in their narratives. By classifying and categorizing visual elements, AI can assist in creating more coherent and immersive science fiction works. The question of the possible development of AI beyond human understanding and control was discussed by Hipple (2020), who investigated works of science fiction with such plots. This research delved into the ethical and existential concerns raised by the potential for AI to surpass human intelligence, exploring themes of autonomy, responsibility, and the implications for human society. This analysis underscores the importance of considering the philosophical and societal impacts of advanced AI technologies.

Freeman and Smith (2023b) argued that deepfakes and AI have erased the gap between 'science' and 'fiction' by combining the contradictions of the genre. Their work examined how AI-generated content, including deepfakes, blurs the lines between reality and fiction, challenging traditional notions of authenticity and truth. This blurring has significant implications for the future of storytelling and the public's trust in media and information. Underwood (2020) emphasized that science fiction has proven to be more stable over two centuries than the fantasy genre. This analysis highlighted the enduring appeal and relevance of science fiction, attributing its stability to its ability to explore and predict technological advancements and societal changes. This stability is contrasted with the more fluid and often ephemeral nature of the fantasy genre, which tends to be more dependent on cultural and temporal contexts.

Research on human interaction with robots and AI, conducted by Dieter and Gessler (2021), focused on how the depiction of technology in the media constructs social reality. Their study explored the influence of media portrayals on public perceptions and attitudes towards robots and AI, emphasizing the role of media in shaping societal norms and expectations. This research underscores the importance of responsible and accurate media representation in fostering a positive and informed relationship between humans and AI. Boucher (2024) critically evaluated the definition of science fiction as the 'literature of the possible', analysing the difference between fantasy and science fiction. This work challenged the traditional boundaries between these genres, arguing that the distinction is often more fluid and complex than commonly acknowledged. By examining the philosophical and literary underpinnings of both genres, Boucher offered a nuanced understanding of their similarities and differences, enriching the discourse on genre classification and literary theory. However, gaps remain that require further study, especially in terms of the impact of these technologies on society and culture.

The researchers immersed themselves in the analysis of powerful lexicographic databases of machine translation systems, but ignored the potential impact of neural network technologies on improving the quality of translations. Their study traced the discussion of the development of artificial intelligence within the limits of human understanding, but has not considered how these technologies can overcome these boundaries and change the landscape of science fiction. Human interaction with robots and artificial intelligence, and their impact on social reality, have remained out of the view of researchers, similar to the psychological and social consequences of using AI, including its impact on cultural processes and personal identity.

The purpose of this study was to investigate the process of introducing artificial intelligence into the creation of fantasy texts. The objectives of this study included several key areas:

1) To explore how the integration of artificial intelligence into the fantasy creation process may have influenced literary creation, and to assess the potential advantages and disadvantages of using AI in this context.
2) To investigate the ethical and legal aspects related to the use of AI for text generation, including copyright and intellectual property issues.

2. Materials and methods

This study employed diverse methodologies to thoroughly examine the influence of artificial intelligence (AI) on literary production, particularly within the fantasy genre. The primary methodologies encompassed literature review, textual analysis, comparative analysis, data analysis, experimental methods, and ethical and legal examination.

Various analytical tools were used to analyse texts and data, including natural language processing

(NLP) software. This allowed assessing the quality and creativity of AI-generated texts on a quantitative level. The first stage of the study included the analysis of literary works created with the help of AI. Several generative models, such as GPT-3 and Gemini, were chosen to create texts. The attention was paid to the comparative analysis of these texts with traditional fantasy works. The comparative analysis was carried out according to key parameters: plot structure, character development, and the use of fantasy elements such as magic and mythical creatures. This helped to identify how well the texts created by AI meet genre standards and how creative they are.

The analysis of texts entailed the comparison and assessment of AI-generated content against human-authored texts. Texts produced by various AI models, including GPT-3 and Gemini AI, were analysed. Special emphasis was placed on the quality and originality of the texts, along with their adherence to genre conventions. Instances of texts are displayed in Tables 1, 2, 3, 4, 5, and 6.

The experimental methods entailed executing experiments with various AI models for text generation and subsequent modification. Comparative experiments were performed using various AI models, including GPT-3 and Gemini AI, to assess their efficacy and the quality of the generated texts. The ethical and legal analyses encompassed an examination of copyright, intellectual property, and liability concerns associated with employing AI to generate literary works. This facilitated the identification of potential risks and issues related to AI utilization, along with providing recommendations for their mitigation. These methods facilitated an extensive examination of the influence of AI on literary creation within the fantasy genre, evaluating its benefits and drawbacks, while addressing the ethical and legal implications of AI utilization in the literature.

3. Results

The fantasy genre has a rich history and has evolved over many centuries. In the modern form, fantasy began its establishment at the beginning of the 20th century, when writers such as J. R .R. Tolkien and K. A. Smith created their own fantasy worlds filled with magic, heroes, and adventures. Over time, the fantasy genre continued to evolve, influenced by new technologies and social changes (Asanov and Turdaliyeva 2024). An important stage was the emergence of computer games and movies, which allowed visualizing fantasy worlds with unprecedented realism and detail. This has opened up new opportunities for authors and artists, allowing them to create not only texts, but also multimedia projects combining various art forms. Modern fantasy not only preserves the classic themes of the struggle between good and evil, magic, and adventure, but also approaches them from new angles, reflecting modern socio-cultural and political realities. It remains a popular genre that continues to inspire and capture readers and viewers with its endless possibilities for fantasy and imagination.

Fantasy originates from mythology, epic poems, and fairy tales such as 'The Epic of Gilgamesh', Greek myths, Scandinavian sagas, and medieval chivalric novels. The main features of the genre, such as magic, mythical creatures and fictional worlds, can be found in these ancient texts (Lebediev 2020). In the second half of the 20th century, the fantasy genre continued to grow and develop. New subgenres have emerged, such as dark fantasy, urban fantasy, and sword and sorcery fantasy. The influence of the genre spread to cinema, television and video games, which contributed to its popularization. In the 21st century, fantasy remains one of the most popular genres, both in literature and in other forms of art (Sokól-Szawlowska and Poleszczyk 2018). Modern authors such as J.K. Rowling with her 'Harry Potter' series, J.R.R. Martin with 'A Song of Ice and Fire' and P. Rothfuss with 'The Name of the Wind', continue to develop the genre by adding new elements and themes (Giuliano 2020).

The fantasy genre, known for its magical worlds, mythological creatures, and epic adventures, provides rich ground for experiments with artificial intelligence (Hennig 2020). The introduction of AI into the process of creating fantasy opens up new opportunities for authors, expanding their creative potential and offering fresh approaches to the development of plots and characters. The use of generative models such as GPT-3 demonstrates the ability of AI to create text based on existing patterns, but its output differs from human creativity and originality. Examples of generated texts include battle scenes and descriptions of fictional worlds that meet genre standards and expected fantasy elements (Table 1).

This example shows that AI can successfully recreate fantasy elements, including magical battles and detailed descriptions of fictional worlds. Using AI to create fantasy works allows creating interactive and personalized worlds where readers can influence the development of the plot and create their own characters. Such interactive platforms make the reading more exciting and engaging (Zhang 2020). An analysis of the perception and experience of authors using AI in their work shows that AI is perceived as a useful tool for overcoming creative blocks and generating new ideas. Many authors note that AI helps to see new directions in the development of the plot, which they would not have thought of on their own, and speeds up the writing process, allowing them to focus on

Table 1. Example of an AI-generated text.

AI-generated battle scene	AI-generated description of the world
In the dark forests of Lorien, ancient forces were preparing for a decisive battle. Elves with drawn swords stood shoulder to shoulder with humans ready to defend their lands. Magical energy was in the air when the sorcerer cast a spell that could change the course of the battle.	The Kingdom of Eldora gleamed in the rays of the setting sun. The tall towers of the castles touched the sky, while life was bustling below. Magicians in their long robes strolled leisurely through the streets, talking about ancient spells and upcoming trials.

Source: compiled by the authors.

Table 2. Example of a paraphrased text with revision.

Source text	AI-edited text
Magicians in their long robes strolled leisurely through the streets, talking about ancient spells and upcoming trials.	The mages, shrouded in long robes, strolled leisurely through the winding streets of the ancient city, animatedly discussing the mysteries of ancient spells and the coming trials that promised great discoveries and dangers.

Source: compiled by the authors.

more complex aspects of creating the world (Hansen 2020; Miller and Sheridan 2024).

The effectiveness of using AI in creating fantasy texts is confirmed by data analysis. Most readers highly appreciate the creativity and coherence of the stories created with the help of AI. This indicates the ability of AI to meet readers' expectations and create high-quality fantasy works (Haroun, Naidu, and Agarwal 2021). Experiments show that generative AI models can not only create coherent texts, but also offer unexpected plot twists, which increases the interest of readers. However, AI-generated texts often require refinement and editing to achieve a high level of quality (Table 2).

The editing shown in Table 2 helps to make the text more vivid and detailed, improving its quality and compliance with genre standards. Artificial intelligence provides significant advantages in the process of creating texts and inspires new creative approaches. It significantly reduces the time required to write large text works, which is especially valuable for authors involved in large-scale projects. Generative AI models help expand creative horizons by offering new ideas and directions for the development of the plot. Moreover, AI allows creating interactive works where readers can influence the course of the plot, which increases their engagement and satisfaction with the reading process. However, the use of AI comes with certain limitations. Texts created using generative models often require significant revision to achieve a high level of quality and give the text liveliness. In addition, questions arise about copyrights and the originality of works, especially in the context of the use of generative algorithms. It is also important to consider ethical aspects, as bias and stereotypes may be embedded in the model, which affects the content created (Agrawal 2023).

Modern technologies significantly influence the development of fantasy literature (Liapis 2021). The introduction of computer technology, the Internet and, of course, AI opens up new opportunities for authors and readers. AI can be used to create more complex and interactive stories that are personalized for each reader. AI capabilities allow authors to experiment with new forms of storytelling and create unique worlds and characters. This contributes to the development of the genre, making it more diverse and dynamic. AI can play an important role at all stages of the creation of fantasy texts, from the initial idea to the final revision. Using AI to generate text can help authors overcome creative blocks and find new ideas for plots and characters. AI can also be used to analyse and process large amounts of data, allowing authors to create more complex and detailed worlds. This is especially important for the fantasy genre, where attention to detail and the logic of the world play a key role. Modern examples of the use of AI in literature include text generation, the creation of interactive stories, and the personalization of content. AI can create texts based on set parameters, which allows authors to focus on other aspects of creativity (Heffernan 2020).

The introduction of AI into the process of creating fantasy texts offers perspectives that until recently seemed fantastic. AI can not only generate texts, but also analyse huge amounts of information, identify trends and preferences of readers, which allows authors to better understand their audience and adapt their works to its needs (Nader *et al.* 2024).

Fantasy characters play a central role, and AI can be useful in developing their characteristics, biographies, and motivations. Based on the analysis of popular fantasy characters, AI can offer original archetypes and detail their behaviour and reactions in various situations (da Silva and Damásio 2022). This helps to avoid cliches and create multifaceted, interesting characters and antagonists. AI can analyse thousands of fantasy plots and identify successful narrative structures, which allows authors to create exciting stories with unexpected twists and tensions. Generative models can offer storylines that inspire authors and help them to avoid routine solutions. As a result, the texts become more dynamic and fascinating for readers.

AI text generation is accompanied by the use of algorithms to check spelling, grammar, and stylistics (Dubourg and Baumard 2021). This allows creating higher-quality texts that require minimal editorial revision. Such technologies are especially useful for aspiring authors, helping them improve their skills and focus on the creative part of the process. AI allows creating personalized texts adapted to the preferences of each reader. This may include the choice of genre elements, the development of the plot according to a certain scenario, or the creation of characters who are close to the reader in spirit. Personalized stories increase the level of engagement and satisfaction of the audience. AI is actively used in creating interactive games and platforms where players can interact with a fictional world and influence the development of the plot. Games like 'The Elder Scrolls' or 'Dragon Age' use sophisticated algorithms to create dynamic worlds and non-linear stories. This allows players to immerse themselves in fantasy worlds and feel part of them. AI is also used to create visual content accompanying fantasy works. This includes the generation of illustrations, maps, and even animated films. The use of AI allows creating high-quality visual materials with minimal time and resources, which is especially important for independent authors and small studios (Zaitseva *et al.* 2023).

The use of AI to create literary works raises questions about the right to authorship and intellectual property. It is important to determine who owns the rights to works created with the help of AI, and how royalties are distributed. These issues require careful study and the development of appropriate legal norms. It is important to develop and apply algorithms that minimize such risks and create inclusive and ethically correct texts. Increasing the use of AI in literature can change the structure of the literary community and affect the profession of a writer. The question arises about the role of human creativity in the context of the active introduction of AI. However, despite possible changes, AI is more likely to become a tool to help authors in their work than a substitute for human creativity. Modern advances in NLP allow AI to better understand and generate texts (Abdrakhmanov *et al.* 2024). This is especially important for creating coherent and creative fantasy plots. NLP can be used to analyse text, identify key topics, and make suggestions for improving structure and style. Developing interfaces that simplify the interaction of authors with AI can make the process of creating works more intuitive and accessible. This may include visual editors, integrated development environments, and collaborative creative tools. Creation of an AI that can adapt to the individual style of the author will allow creating more unique and personalized works. Such models can be trained on the texts of a particular author, which allows preserving their style and intonation in the generated texts.

Continuing research in the field of ethical and legal aspects of the use of AI in the literature will help resolve issues related to copyright, intellectual property, and liability. This is necessary to create a fair and transparent environment for all participants in the literary process. Journalism professor Shen Yang used artificial intelligence to create a science fiction work that won a prize at the national science fiction competition for young authors. The chatbot developed the idea, text, illustrations, and even the writer's pseudonym. Shen used only 66 queries to create a small novel called 'The Land of Memories', but independently finalized the drafts provided by the AI platform. This story of a journey through the metaverse, according to the jury of the festival, resembles the works of Kafka. The plot of the novel revolves around the developer of artificial intelligence L. Xiao, who loses the memory of her family as a result of a failed experiment. After learning about the legend of the Land of Memories, she decides to go there to regain what she has lost. In this region, she encounters robot hallucinations and illusions created by AI (Brovinska 2022).

The novel was presented at a youth prose competition organized by the Science Fiction Union of Jiangsu Province, and took second place, receiving the votes of three of the six jury members. Only one of the judges knew that the work was created by AI, and the other guessed it after carefully studying the text, considering it devoid of emotions and not meeting the selection criteria. Shen said that he deliberately asked AI to write a book in the style of Franz Kafka, and after a dozen requests, the chatbot satisfied all his requirements. The professor plans to describe in detail the entire process of working on the book and post it on the Internet to show how science fiction writers can use neural networks. Fu Ruchu, editor-in-chief of People's Literature Publishing House, said it was difficult for her and her colleagues to guess that the piece was not written by a human. Despite the laconic and emotionless language of the book, which is typical for many Chinese science fiction writers, Fu fears that with the growing popularity of AI, authors may lose their sense of language, which is already rare (Haydamashko 2023). In the middle of March 2022, J. Lepp was in the final stages of work on the book ‘Bring Your Beach Owl’, but was significantly behind schedule (Zhu and Zhang 2022). Her spreadsheet showed that she had three days left to write 9,278 words and prepare the book for publication on Amazon and Kindle. J. Lepp became a writer six years ago after leaving her corporate job. She uses Sudowrite, an AI tool, to improve her texts. The programme helped her to create vivid descriptions and inspired her with new ideas. However, she realized that it was not worth relying on AI completely, but using it as a source of inspiration.

Language models such as GPT-3 work like predictive word machines. After receiving a huge amount of texts, they adjust their billions of initially random parameters to predict the next words. Due to the large amount of text and a variety of parameters, the GPT-3 can perform basic arithmetic operations, translate languages and write working code—even without learning these skills. A researcher under the pseudonym G. Branwen calls this ‘fast programming’. For example, GPT-3 can write in the style of certain authors if an initial text is provided. Sudowrite, based on GPT-3, helps authors to improve their texts by creating descriptions, plot twists, and metaphors. Independent writer Joanna Penn suggests that authors should use AI to expand their capabilities and improve their works. She helps authors master AI and has developed a code of ethical conduct for working with AI. However, not everyone shares her enthusiasm. Some consider the use of AI to be cheating. Penn and J. Lepp suggest that AI is just another tool to help authors create texts (Zhu and Zhang 2022). J. Lepp strives to preserve the ‘crazy’ unpredictability of her texts, even if this means recycling a significant amount of unnecessary material. She likes that the hallucinatory nature of AI directs her in unexpected directions, but the control of the story remains with her.

Working with AI, J. Lepp got into a rhythm, allowing AI to write, and then made edits. This allowed her to increase productivity by 23.1 percent. J. Lepp uses Sudowrite less frequently, providing the programme with incomplete sentences or the basis of a scene. She notes that this reduces the emotional burden and makes the process less tedious, keeping in touch with the story. She is currently working on two series at the same time, planning to release 10 books a year. Nevertheless, J. Lepp understands the fears of colleagues that authors will not be able to compete with AI in writing speed (Zhu and Zhang 2022). Authors interested in automated writing can contact D. Rollins, who helps create books for various businesses using AI. Jasper.ai used by D. Rollins, solves text length problems using templates, allowing AI to write more coherent and structured texts (Table 3).

Analysing Table 3, it can be highlighted that the text has a classic fantasy genre structure: the hero goes in search of an artefact that can change the course of history, and encounters an ancient evil. The plot develops dynamically, with an emphasis on magic and ancient prophecies, which is typical for fantasy. The text is written in a narrative style with elements of description and dialogue. Vivid and detailed descriptions are used to help create an atmosphere of the magical world. The language is simple but expressive, which makes the text accessible to a wide range of readers. The main character, Elian, is presented as a young magician seeking to save the world. He has determination and courage, but is also aware of his responsibility for awakening evil. This makes him a character that readers can easily empathize with. The main themes of the text include the struggle between good and evil, the power of magic, and responsibility for own actions. The motifs of ancient prophecies and magical artefacts add depth and intrigue to the plot. Descriptions of a magical forest and an abandoned temple create vivid visual images that help readers immerse themselves in the fantasy world. The use of magical symbols and glowing runes adds a mystical atmosphere. The text can be adapted for interactive platforms where readers could influence Elian’s choices and the development of the plot. This would increase the engagement and satisfaction of the audience.

The ‘Wars of the Forgotten Kingdoms’ demonstrates how artificial intelligence can create creative and fascinating fantasy texts. Despite the presence of some stylistic and logical errors, the text demonstrates a high degree of creativity and originality. The integration of AI into the process of creating fantasy works opens up new perspectives for the genre and makes it more accessible and interesting to a wide audience (Duca 2024). Next, the artificial intelligence was asked to rewrite its own fantasy text, but in the style of the writer S.J. Maas (Table 4).

The text retains the classic fantasy structure: the hero goes in search of an artefact, encounters an ancient evil, and understands the need to unite forces to win. The plot is dynamic and eventful, which is typical of the style of S.J. Maas. The text is written in a vivid, emotional, and detailed style, which is inherent in S. J. Maas. Rich descriptions and internal monologues are used, which allows a deeper understanding of the main character’s experiences and motives. The main character, Elian, is presented as a complex and multifaceted character. His determination, courage, and awareness

Table 3. Example of a fantasy text created using AI.

Name	Wars of the Forgotten Kingdoms
Abstract	In a distant magical world where ancient kingdoms are fighting for power, a young magician named Elian finds an artefact that can change the course of history. However, along with a powerful force, the artefact also brings great evil that threatens to destroy the whole world. Elian and his companions must go through many trials to prevent an impending catastrophe.
Chapter 1: The Mystery of the Artefact	In the heart of the ancient forest, under the cover of night, Elian walked along a narrow path leading to an abandoned temple. Legends said that in its depths there was a powerful artefact that could change the fate of the world. The magicians who lived in this kingdom had been searching for it for a long time, hoping to gain the power to protect their lands from enemies. When Elian entered the temple, he felt a powerful magical field. The walls were decorated with ancient runes glowing with a dim blue light. In the centre of the hall stood an altar, on which lay an artefact—a sparkling crystal surrounded by magical symbols. 'Here it is,' Elian whispered, reaching for the crystal. – 'The power that the legends talked about'. But as soon as he touched the artefact, a bright light flashed around. Elian felt something dark and sinister awaken in the depths of the temple. He knew that along with the force, he had also freed the great evil that the ancient prophecies had warned about.
Chapter 2: The Awakening of Evil	Elian, holding the artefact in his hands, felt the power filling his body. But with this power came the awareness of danger. Dark creatures rose from the depths of the temple, their eyes burning with malice and hatred. 'You shouldn't have awakened us', one of the creatures hissed, approaching Elian. 'Now the whole world will pay for your mistake'. Elian gritted his teeth and grabbed the artefact. He knew that his only chance was to use the power of the crystal to contain the evil. 'I will not let you destroy our world', he said, directing magical energy at the dark creatures. But the evil was too powerful. Elian realized that he would need help to defeat these dark forces and bring peace back to the Forgotten Kingdoms.

Source: compiled by the authors.

of his responsibility to the world make him attractive to readers. Secondary characters and creatures add depth and intrigue, which is also typical of S.J. Maas's style.

The work 'Wars of the Forgotten Kingdoms' in the style of S.J. Maas demonstrates how AI can create creative and emotionally intense fantasy texts. Vivid descriptions, complex characters, and a dynamic plot make the text interesting and fascinating. The integration of AI into the process of creating fantasy works opens up new perspectives for the genre and makes it more accessible and attractive to a wide audience. Next, a text in the fantasy genre was generated using Gemini AI (Table 5). It is important to point out that Gemini AI, unlike GPT3-5, asked to detail the fantasy story with various questions about the type of magic, the type of world, the conflict in it, and the description of the main plot.

After Gemini rewrote this text in the style of S. J. Maas and pointed out that 'the style of S.J. Maas is distinguished by vivid descriptions, emotionality, deep immersion into the inner world of the characters, and a romantic line that is closely intertwined with the plot (Table 6). Her prose is often compared to poetry, due to colourful metaphors and epithets'.

Instead of a simple description of the area and events, vivid metaphors and comparisons were used to convey the atmosphere and feelings of the characters. There was more emphasis on Ellania's inner feelings, emphasizing her loneliness and fear. The romantic relationship between Ellania and Nora is presented as something special that can inspire feats. The style of S. J. Maas allows the reader to delve deeper into the story, feel the emotions of the characters, and enjoy the beauty of the descriptions. The reworking of the text in her style made it more emotional and poetic, while preserving the main storyline. In conclusion, the study highlighted the importance of the fantasy genre as a means of cultural and social expression, and the prospects for further development of the genre in the context of new technologies. The integration of AI into literary creativity opens up new opportunities for

Table 4. A fantasy text rewritten in the style of S. J. Maas.

Name	Wars of the Forgotten Kingdoms
Abstract.	In a world full of magic and ancient secrets, a young magician named Elian discovered a powerful artefact that can change the course of history. But along with his power, an ancient evil awakens, threatening to destroy everything he loves. Elian and his companions must go through many trials to prevent an impending catastrophe.
Chapter 1: The Mystery of the Artefact	Night fell on the ancient forest, shrouding it in darkness and mystery. Elian carefully walked along a narrow path leading to an abandoned temple hidden in the depths of the forest. Legends said that there was an artefact hidden in this temple whose power could change the fate of the world. Magicians with ancient knowledge have been searching for it for centuries, hoping to find a force capable of protecting their kingdoms from danger. When Elian entered the temple, he was surrounded by a mystical light emanating from the walls covered with ancient runes. In the centre of the hall stood an altar on which rested a sparkling crystal surrounded by magical symbols. 'Here it is'.—Elian whispered, his heart beating faster.—'The power that the legends talked about'. His fingers touched the crystal lightly, and the world exploded with bright light. Elian felt an unknown force awakening in him, but with it something dark and sinister. He knew that he had awakened something that would have been better left in oblivion.
Chapter 2: The Awakening of Evil	The power of the artefact filled Elian's body, but with it came the realization of horror. Creatures whose eyes glowed with hatred began to rise from the dark depths of the temple. – 'You shouldn't have awakened us', one of the creatures hissed, approaching Elian. 'Now the whole world will pay for your mistake'. Elian gritted his teeth as he picked up the artefact. He knew that his only chance was to use the power of the crystal to contain the evil. 'I won't let you destroy our world', he said, directing magical energy at the dark figures. But their strength was too great. Elian realized that he would need help to defeat these creatures and bring peace back to the Forgotten Realms.

Source: compiled by the authors.

authors, making the process of creating works more dynamic and innovative.

4. Discussion

The analysis of the findings shows the significant influence of artificial intelligence on the process of creating works in the fantasy genre. First of all, it is obvious that AI provides authors with a powerful tool to improve the quality and speed up the writing process. This is becoming especially important in today's market, where the speed and quality of content production are of key importance. The results of the study emphasize that the use of AI can significantly expand the possibilities of visualization of fantasy worlds. Computer graphics and AI algorithms are capable of creating complex and detailed images, which increases the attractiveness of the genre and contributes to its popularization (Oliinyk *et al.* 2023). Interactive AI-based platforms provide readers with the opportunity to actively participate in plot creation and develop their own characters, which enhances audience engagement and satisfaction (Takovski 2020). The study also revealed that modern fantasy works created using AI retain the basic elements of the genre, such as the struggle between good and evil, magic, and adventure. However, due to AI, these works can include modern socio-cultural and political themes, which makes them more relevant and interesting for a wide audience. The integration of AI allows authors to more effectively overcome creative blocks and generate new ideas, which contributes to the creation of more diverse and creative stories.

The ethical and legal aspects of the use of AI in literature are increasingly important areas of research. Copyright and intellectual property issues are particularly relevant in the context of creating texts using AI. As AI becomes more integrated into the literary process, there is a growing need to develop new regulations and standards that ensure the protection of authors' rights and govern the use of AI in writing. Varsha (2023) highlights the complexities of managing

Table 5. Fantasy story generated using AI.

Requests	Generated text
World: Dark and gloomy. Protagonist: Princess. Conflict: Political intrigue, struggle for power. Magic: magic has been hidden from the general population for a long time, but due to the conflict in secret societies, it becomes evident, Romance: there is a romantic line, tropes from enemies to lovers. A young princess hiding her magical abilities is forced to flee her native kingdom after a conspiracy. She finds herself in a world where magic is forbidden, and she has to fight for her freedom and save her family.	In the war-torn kingdom of Ethel, where shadows curled through the cobbled streets of Twilight Harbour and ancient magic awoke in the depths of Whispering Forests, lived Princess Ellania Stormbringer. The heir to a crumbling throne, she kept a secret—the ability to wield illusions. The Obsidian Hand, a secret society of magicians, sought to use the power of Ellania Stormbringer for their insidious purposes. However, their plans collapsed, and Ellania Stormbringer was forced to flee, pursued by those who had once promised to protect her. During her escape, Ellania Stormbringer found refuge in the most unexpected place—the Whispering Forests. There she met Nora Stormbringer, a highwaywoman with a difficult past. Despite the initial distrust, a bond developed between them, blossoming into a love that defies expectations. Fueled by a thirst for revenge and a desire to restore peace, Ellania and Nora Stormbringer have assembled a band of rogue rebels, disillusioned soldiers and magicians eager for a world free from the tyranny of the Obsidian Hand. A power struggle was brewing inside the Obsidian Hand. Moderates, tired of the bloodshed, faced radicals obsessed with domination. Ellania saw an opportunity. She infiltrated society by using her intelligence and charm to manipulate factions against each other. The trials she faced hardened Ellania Stormbringer. The once-pampered princess has turned into a formidable warrior, her magic has become a weapon honed by desperation and love. She learned the true meaning of leadership by inspiring loyalty and courage in those who followed her. The inevitable clash has come. In an exciting duel, Ellania Stormbringer and her allies faced off against the combined might of the Obsidian Hand. The battle raged, magic crackling in the air as steel met steel. Sacrifices were made, but in the end, due to a cunning strategy and unwavering determination, Ellania Stormbringer won. With the defeat of the Obsidian Hand, a new morning dawned over Ethel. Ellania Stormbringer, proclaimed the liberator, ushered in an era of peace and prosperity. But the scars of the war remained, a constant reminder of the sacrifices made. Nevertheless, Nora Stormbringer stood next to her, a testament to the enduring power of love and rebellion.

Source: compiled by the authors.

biases in AI systems, which extends to the ethical considerations of AI-generated texts. Similarly, Bamboat and Yu (2023) discuss the need for fairness in machine learning systems, emphasizing the importance of ethical guidelines in the development and deployment of AI.

Concerns that AI can replace human creativity remain relevant, and texts created by AI are often criticized for lack of creativity and depth. Pathak *et al.* (2024) explore how gender bias in AI-based developments can lead to discriminatory outcomes, underscoring the need for inclusive datasets and bias mitigation techniques. Despite these limitations, the results of the study show that the use of AI opens up new perspectives for the fantasy genre. AI can analyse large amounts of data and identify trends and preferences of readers, allowing authors to more accurately adapt their works to the needs of the audience. This makes the fantasy genre more dynamic and able to meet the challenges of the time, while remaining popular among various groups of readers. By addressing the ethical and legal aspects, and leveraging the

Table 6. Fantasy story paraphrased using AI.

A rewritten fragment of the story in the style of S. J. Maas
In Ethel's realm, where shadows laced the cobbled streets of Twilight Harbour and ancient magic pulsed through the veins of Whispering Forests, lived Princess Ellania Stormbringer. Like the night sky hiding the abyss of its secrets, she kept in her heart the gift of illusions, flickering with a spark of forbidden magic.
The Obsidian Hand, the shadow brotherhood of magicians, was eager to get this spark, turn it into a weapon of domination. But fate decreed otherwise. Haunted by the shadows of betrayal, Ellania was forced to flee like a moth escaping from a flame.
In the depths of the Whispering Forests, where the sun's rays pierced through the foliage like golden arrows, she found shelter. There, among the ancient trees, her path crossed with Nora Stormbringer, a robber with raven eyes and a heart full of scars. A spark flared up between them, timid and gentle, like the first flower of spring, but capable of igniting the flames of rebellion.

Source: compiled by the authors.

capabilities of AI, the literary world can evolve to be more inclusive, innovative, and responsive to the diverse needs of its audience.

Research by Zhu and Zhang (2022) on films analyses the impact of AI on the film industry, dividing the genre into two levels: technical practicality and plot creation. It emphasizes that with the development of science and technology, the orientations of creating AI films are changing, reflecting the real world and at the same time influencing it. This study considers AI film as a means that can inspire real world developments and change the perception of the audience. Research by Zhu and Zhang (2022) and this study have common features in analysing the impact of AI on creativity. Both studies emphasize that AI significantly influences the process of creating content, be it films or literary works, improving their quality and speeding up production. Both studies note that texts created using AI can reflect and interpret contemporary sociocultural and political themes, thereby making them more relevant. Unlike the study by Thorne (2020), which focuses on improving and enriching the genre through AI, this study focuses on commercial applications of AI and its ability to completely replace human authors. Research by Zhu and Zhang (2022) and Thorne (2020) has several key similarities and differences. Both studies examine the use of AI in various forms of creativity, such as movies, games, and literature, highlighting its impact on the content creation process. The conducted research focuses on the commercial aspects and the possibilities of completely replacing human authors with algorithmically generated stories, whereas the study by the researcher is focused on improving and enriching the fantasy genre through AI. Unlike the study by Thorne (2020), which considers AI as a tool to support and empower authors, this study focuses on the ability of AI to completely replace human authors.

Baiheng and Wen (2020) focus on the intervention of artificial intelligence agents in narratology and on building a generalized model of the world narrative. The main goal of this research is to expand AI stories to electronic literature and attract the attention of researchers from various fields such as computer science, communication, art, design, and psychology. It is expected that this will attract interest and lead to numerous experiments and research, which should contribute to the development of this field. Baiheng and Wen (2020) and the conducted research both focus on the use of artificial intelligence in the content creation process, but do so from different perspectives and goals. The main purpose of the study by the researchers aims to expand the application of AI stories to electronic literature and attract the attention of a wide range of researchers, while research on the fantasy genre and artificial intelligence focuses on how AI can improve the process of creating fantasy works, making them more interactive and personalized.

The paper by Danesi (2024) analyses the impact of advanced artificial intelligence technologies on rethinking the concepts of intelligence and creativity. It emphasizes that AI systems are capable of creating works of art, music, films, videos and advertisements that are perceived positively, win awards, and go viral on social networks. The study examines the consequences of this event from the standpoint of semiotics as the science of creating meanings, emphasizing the need to rethink the concepts of authenticity and authorship, including traditional views on creativity, culture, and human uniqueness. Both studies examine the impact of advanced AI technologies on the creative process, whether it is the creation of works of art, music, films, or literature. Both studies emphasize the need to rethink traditional concepts such as authorship, creativity, and human uniqueness in light of the development of AI. The author examines the consequences of using AI from the standpoint of semiotics, the science of creating meanings, whereas research on the fantasy and AI genre is more focused on the practical application of AI to improve the quality and interactivity of fantasy works.

Boucher (2024) considers fantasy literature as a study of subjectivity and the literature of the impossible. He offers a representative definition of fiction and a modal approach to its core set. Boucher (2024) argues that the marvellous mode includes other speculative literatures, the difference of which is in the reference world. He distinguishes between fiction as a literature of speculative subjectivity, governed by an 'affective novum', and science fiction as a literature of conjunctural objectivity, governed by a 'cognitive novum'. In comparison with the research conducted above, which focuses on the integration of AI into the creation of fantasy texts and their practical use, the study by the researcher offers a theoretical analysis of genre differences.

The study by Dear (2023) analyses the theological concept of imago Dei in the context of artificial intelligence using the example of Kazuo Ishiguro's novel 'Clara and the Sun'. It shows that fiction is useful for exploring hypothetical scenarios, including the emergence of strong AI, due to the ability to consider untested realities. Text analysis reveals difficulties in considering human uniqueness in the context of AI. The substantial model is rejected due to the superior intellectual abilities of AI in the novel, emphasizing the differences between human and machine thinking. Clara, the narrator of the novel and AI, fits into a functional interpretation, reflecting concerns about the exploitation of nature and the replacement of man. A study on the integration of artificial intelligence into fantasy creation, and the paper by Dear (2023), analysing the theological concept of imago Dei in the context of AI, present two different approaches to the study of the role and influence of AI in literature. The research conducted above focuses on the practical use of AI in the creative process. It shows how AI empowers authors by providing them with new tools to overcome creative blocks and generate ideas. Visualization technologies improved by AI contribute to the popularization of the genre, making fantasy worlds more realistic and attractive (Maltsev *et al.* 2022). In contrast, the study by Dear (2023) uses Kazuo Ishiguro's novel 'Clara and the Sun' to analyse theological and philosophical issues related to AI. The study shows how fiction can be useful for exploring hypothetical scenarios, such as the emergence of strong AI. Ishiguro's novel raises questions about human uniqueness in the context of AI, which is especially evident through the analysis of substantial and functional models. In the novel, AI's superior intellectual abilities highlight the differences between human and machine thinking, and narrator Clara presents a functional interpretation, reflecting concerns about the exploitation of nature and the replacement of man.

Burton (2022) examines the depictions of AI in literature and media for young people to understand how their perception of AI and related ontological and social issues are shaped through interaction with young people. It analyses why the perception of AI can be so different, comparing frightening images such as Terminator and HAL 9000 with friendly images such as Pinocchio and Wall-E. The study uses posthumanistic theory, which destroys traditional categories and hierarchies, offering analysis through the prism of hybridity and intra-action. In comparison with the study conducted above, which focuses on the integration of AI into the process of creating fantasy works and assessing its impact on the genre, the current study focuses on the cultural and philosophical perception of AI in the context of youth literature and media. It examines how AI imagery affects the understanding and perception of technology and its place in society, while the research from the file focuses on the practical application of AI to enhance visualization and interaction in the fantasy genre. Thus, the main difference between these research works lies in their focus: the study above focuses on the practical application of AI in fantasy creation, while the current study explores the cultural perception of AI through youth literature and media.

5. Conclusions

The study concludes that AI is a powerful tool that significantly expands the capabilities of authors and readers. It helps to speed up the writing process, improve the quality of texts, and offer new forms of interaction with content. The use of AI in literature is becoming an important aspect of modern creativity, providing authors with new means to implement their ideas. In addition, modern technologies such as computer graphics and AI have significantly expanded the possibilities of visualizing fantasy worlds. This not only increased the popularity of the genre, but also allowed creating deeper and richer texts that can attract and retain the attention of a wide audience. Interactive platforms and personalized worlds created using AI increase the engagement and satisfaction of readers, offering them a unique experience of interacting with the work.

In addition, the use of AI in the creation of fantasy works opens up new perspectives for the genre. AI can not only generate texts, but also analyse huge amounts of information, identify trends and preferences of readers, which allows authors to better understand their audience and adapt their works to its needs. This makes the fantasy genre more diverse and dynamic, able to meet the challenges of the time and remain popular among readers of different ages and interests.

The use of artificial intelligence in the creation of fantasy works is a promising area that opens up new opportunities for authors and readers.

Author contributions

Yertay Sultan (Conceptualization, Formal analysis, Methodology, Writing–original draft), Gulnaz Dautova (Conceptualization, Investigation, Methodology, Writing–review & editing), Dina Alkebayeva (Data curation, Formal analysis, Writing–review & editing), Akkibat Akzhigitova (Funding acquisition, Investigation, Supervision, Writing–original draft), Zhansaya Aden (Formal analysis, Project administration, Writing–review & editing).

Funding

This research was funded by the Science Committee of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant No. AP23489843 'Foundations of Artificial Intelligence in Literature: Unveiling Core Mechanisms, Implications, and Future Trajectories').

Data availability

The data underlying this article are already available within the article.

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